

ELEC 3200: System Modeling, Analysis and Control

Lecture 18: Second-Order Systems

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Summary

$$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s + \sigma)^2 + \omega_d^2}$$

$$\omega_n = \sqrt{A_0}, \quad \zeta = \frac{A_1}{2\omega_n} = \frac{A_1}{2\sqrt{A_0}}$$

$$s = -\omega_n \zeta \pm \omega_n \sqrt{\zeta^2 - 1} = -\sigma \pm j\omega_d$$

$$\sigma = \zeta\omega_n, \quad \omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$\zeta \leq 0 \Rightarrow$ unstable system

$|\zeta| < 1 \Rightarrow$ oscillatory system

$\zeta \gg 1 \Rightarrow$ over damped

$\zeta \approx 1 \Rightarrow$ critically damped

$\zeta \rightarrow 0^+ \Rightarrow$ under damped

$$t_r = \frac{1.8}{\omega_n}$$

$$t_p = \frac{\pi}{\omega_d}$$

$$M_p = \exp\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

$$t_s = \frac{3}{\sigma}$$

Prototype 2nd-Order System

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

By the quadratic formula, the poles are:

$$\begin{aligned} s &= -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \\ &= -\omega_n \left(\zeta \pm \sqrt{\zeta^2 - 1} \right) \end{aligned}$$

The nature of the poles changes depending on ζ :

- ▶ $\zeta > 1$ both poles are real and negative
- ▶ $\zeta = 1$ one negative pole
- ▶ $\zeta < 1$ two complex poles with negative real parts

$$s = -\sigma \pm j\omega_d$$

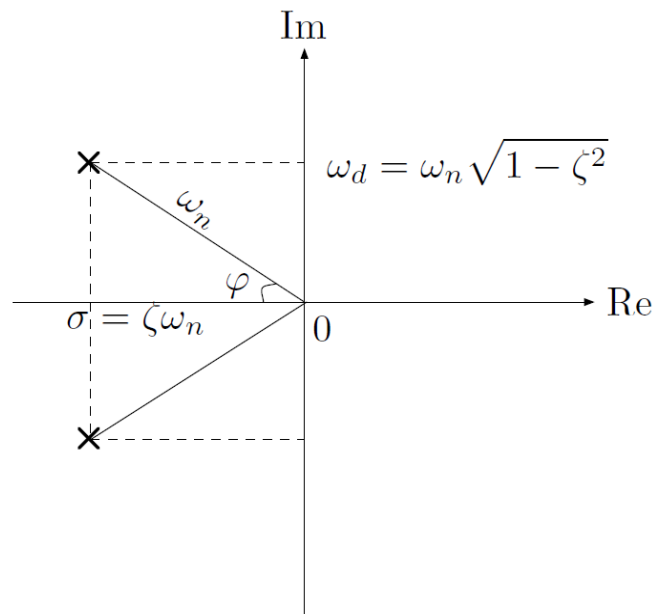
$$\text{where } \sigma = \zeta\omega_n, \omega_d = \omega_n\sqrt{1 - \zeta^2}$$

Prototype 2nd-Order System

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad \zeta < 1$$

The poles are

$$s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -\sigma \pm j\omega_d$$



Note that

$$\begin{aligned} \sigma^2 + \omega_d^2 &= \zeta^2\omega_n^2 + \omega_n^2 - \zeta^2\omega_n^2 \\ &= \omega_n^2 \end{aligned}$$

$$\cos \varphi = \frac{\zeta\omega_n}{\omega_n} = \zeta$$

2nd-Order Response

Let's compute the system's impulse and step response:

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s + \sigma)^2 + \omega_d^2}$$

► Impulse response:

$$\begin{aligned} h(t) &= \mathcal{L}^{-1}\{H(s)\} = \mathcal{L}^{-1}\left\{\frac{(\omega_n^2/\omega_d)\omega_d}{(s + \sigma)^2 + \omega_d^2}\right\} \\ &= \frac{\omega_n^2}{\omega_d} e^{-\sigma t} \sin(\omega_d t) \end{aligned}$$

► Step response:

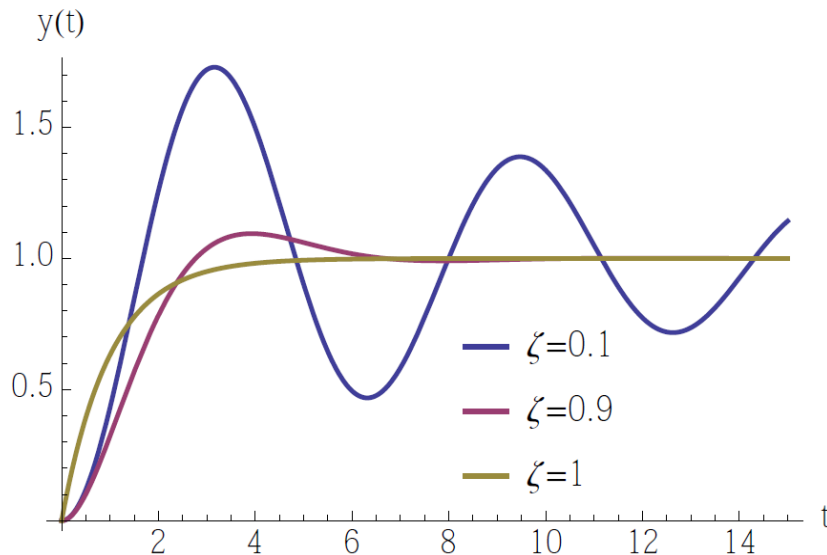
$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{H(s)}{s}\right\} &= \mathcal{L}^{-1}\left\{\frac{\sigma^2 + \omega_d^2}{s[(s + \sigma)^2 + \omega_d^2]}\right\} \\ &= 1 - e^{-\sigma t} \left(\cos(\omega_d t) + \frac{\sigma}{\omega_d} \sin(\omega_d t) \right) \quad \text{(Proof is given in the appendix)} \end{aligned}$$

2nd-Order Step Response

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s + \sigma)^2 + \omega_d^2}$$

$$u(t) = 1(t) \quad \longrightarrow \quad y(t) = 1 - e^{-\sigma t} \left(\cos(\omega_d t) + \frac{\sigma}{\omega_d} \sin(\omega_d t) \right)$$

where $\sigma = \zeta\omega_n$ and $\omega_d = \omega_n\sqrt{1 - \zeta^2}$ (damped frequency)



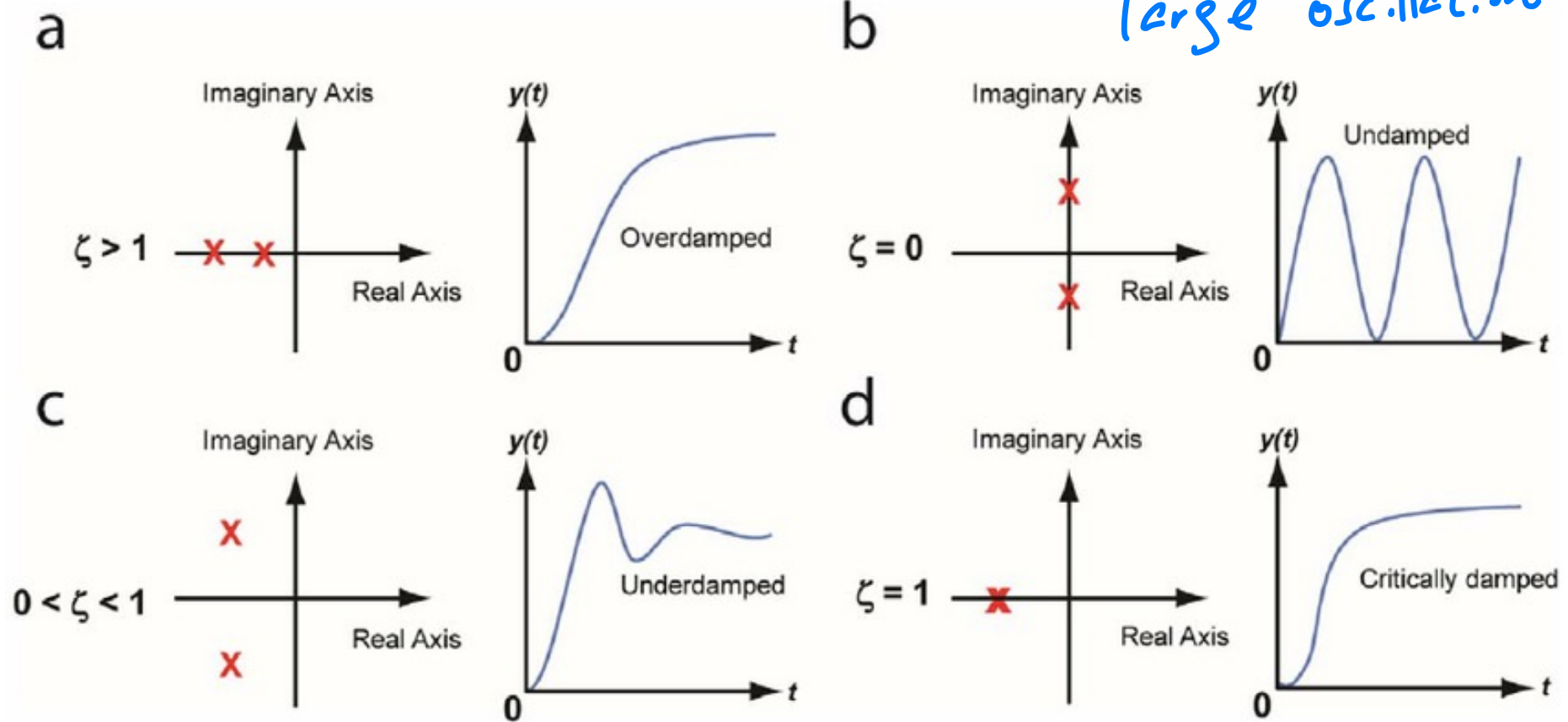
The parameter ζ is called the *damping ratio*

- ▶ $\zeta > 1$: system is overdamped
- ▶ $\zeta < 1$: system is underdamped
- ▶ $\zeta = 0$: no damping ($\omega_d = \omega_n$)

2nd-Order Step Response

$$s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -\sigma \pm j\omega_d$$

large oscillation



2nd-Order Step Response

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s + \sigma)^2 + \omega_d^2}$$
$$u(t) = 1(t) \quad \longrightarrow \quad y(t) = 1 - e^{-\sigma t} \left(\cos(\omega_d t) + \frac{\sigma}{\omega_d} \sin(\omega_d t) \right)$$

where $\sigma = \zeta\omega_n$ and $\omega_d = \omega_n\sqrt{1 - \zeta^2}$ (damped frequency)

We will see that the parameters ζ and ω_n determine certain important features of the transient part of the above step response.

We will also learn how to pick ζ and ω_n in order to *shape* these features according to given *specifications*.

First see first order system!

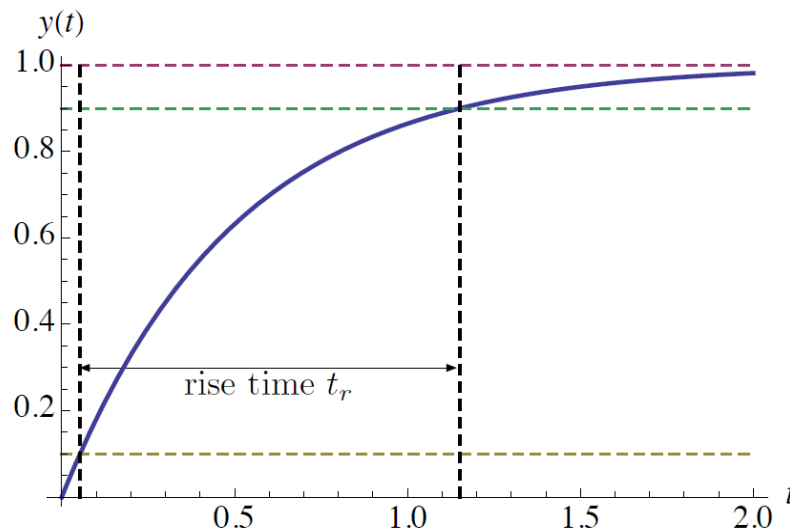
Transient Response Specifications: Rise Time

Let's first take a look at *1st-order step response*

$$H(s) = \frac{a}{s + a}, \quad a > 0 \quad (\text{stable pole})$$

DC gain = 1 (by FVT)

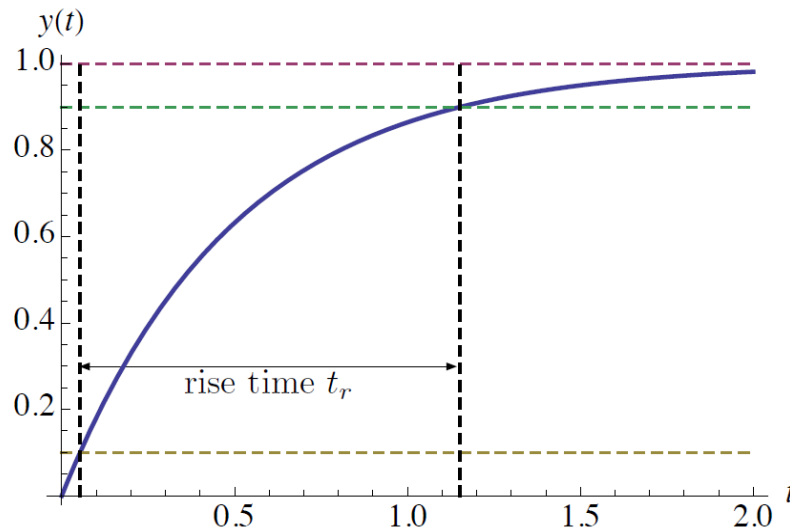
Step response:
$$Y(s) = \frac{H(s)}{s} = \frac{a}{s(s + a)} = \frac{1}{s} - \frac{1}{s + a}$$
$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = 1(t) - e^{-at}$$



Rise time t_r : the time it takes to get from 10% of steady-state value to 90%

Rise Time

Step response: $y(t) = 1(t) - e^{-at}$



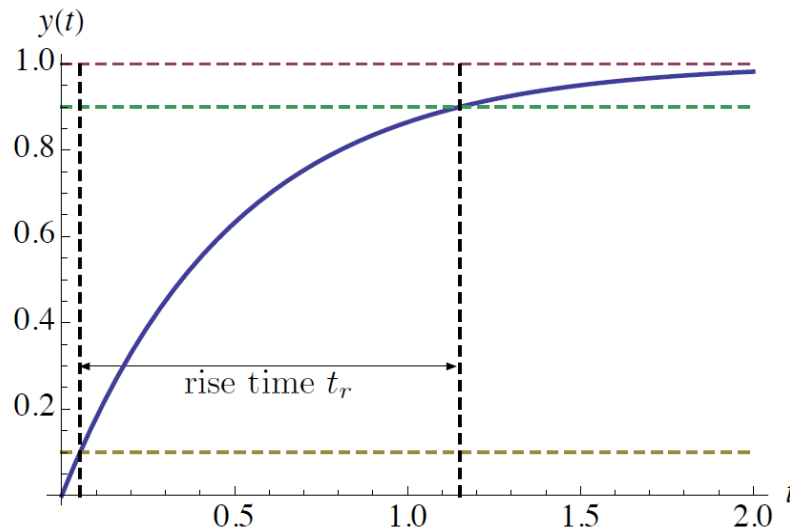
Rise time t_r : the time it takes to get from 10% of steady-state value to 90%

Examples of rise time:

- ▶ car — going from 0 to 60 mph in 7 sec
- ▶ oven — reach desired preheat temperature quickly
- ▶ thermostat, building climate control
- ▶ other examples?

Rise Time

Step response: $y(t) = 1(t) - e^{-at}$



Rise time t_r : the time it takes to get from 10% of steady-state value to 90%

In this example, it is easy to compute t_r analytically:

$$1 - e^{-at_{0.1}} = 0.1 \quad e^{-at_{0.1}} = 0.9 \quad t_{0.1} = -\frac{\ln 0.9}{a}$$

$$1 - e^{-at_{0.9}} = 0.9 \quad e^{-at_{0.9}} = 0.1 \quad t_{0.9} = -\frac{\ln 0.1}{a}$$

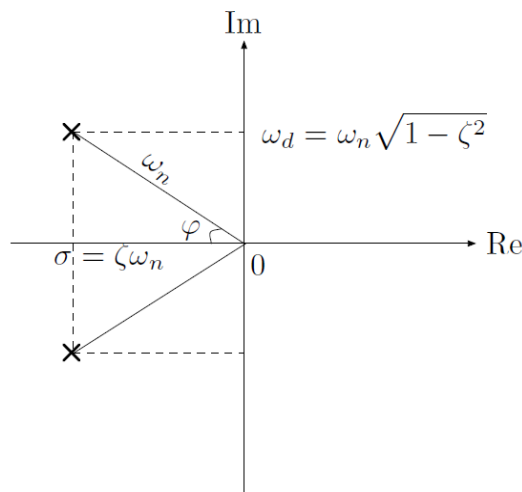
$$t_r = t_{0.9} - t_{0.1} = \frac{\ln 0.9 - \ln 0.1}{a} = \frac{\ln 9}{a} \approx \frac{2.2}{a}$$

Transient Response Specs

Now let's consider the more interesting case: *2nd-order response*

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s + \sigma)^2 + \omega_d^2}$$

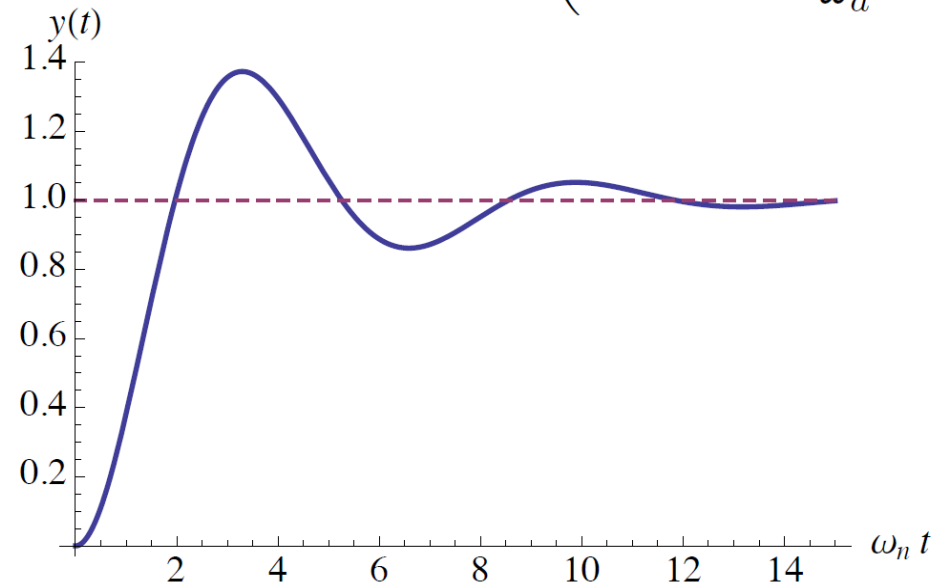
where $\sigma = \zeta\omega_n$ $\omega_d = \omega_n\sqrt{1 - \zeta^2}$ ($\zeta < 1$)



Step response: $y(t) = 1 - e^{-\sigma t} \left(\cos(\omega_d t) + \frac{\sigma}{\omega_d} \sin(\omega_d t) \right)$

Transient-Response Specs

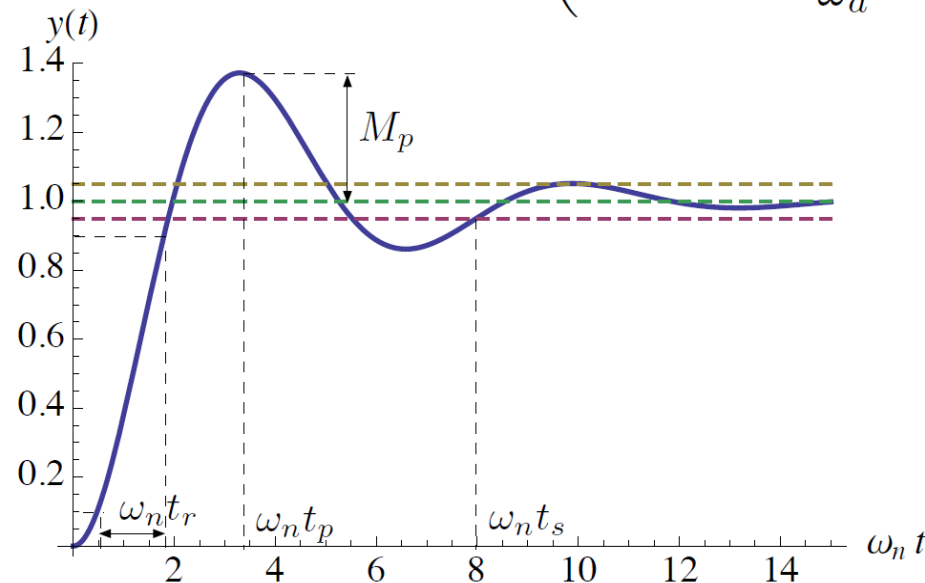
Step response:
$$y(t) = 1 - e^{-\sigma t} \left(\cos(\omega_d t) + \frac{\sigma}{\omega_d} \sin(\omega_d t) \right)$$



- rise time t_r — time to get from $0.1y(\infty)$ to $0.9y(\infty)$
- overshoot M_p and peak time t_p
- settling time t_s — first time for transients to decay to within a specified small percentage of $y(\infty)$ and stay in that range (we will usually worry about 5% settling time)

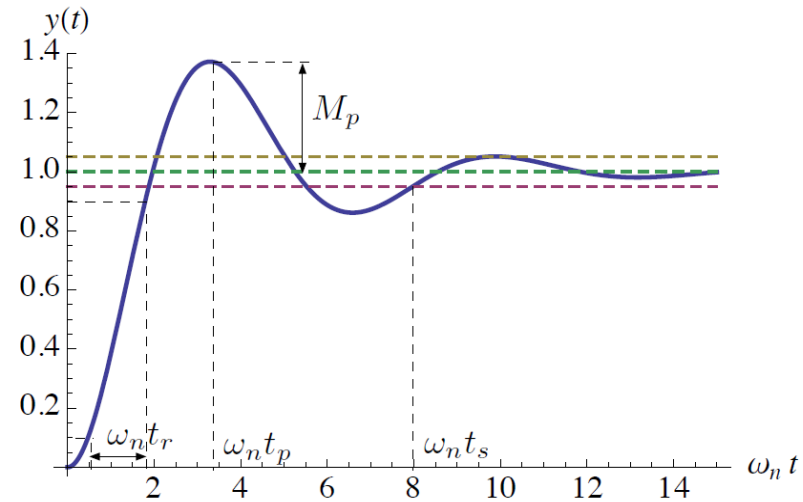
Transient-Response Specs

Step response: $y(t) = 1 - e^{-\sigma t} \left(\cos(\omega_d t) + \frac{\sigma}{\omega_d} \sin(\omega_d t) \right)$



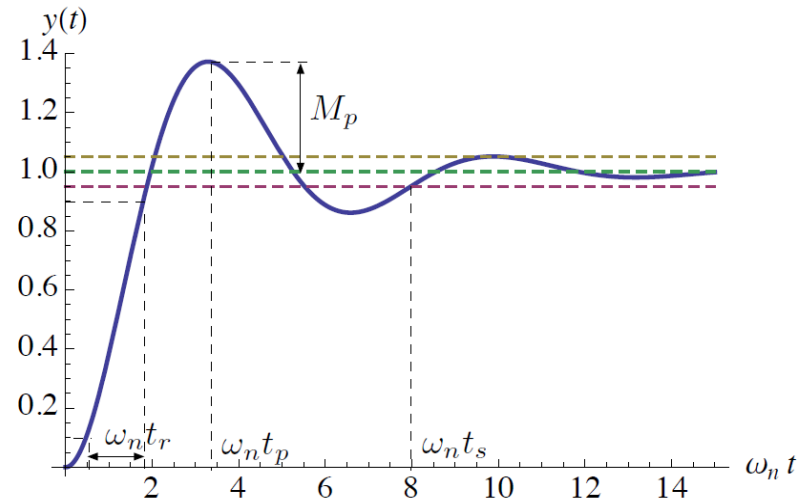
- rise time t_r — time to get from $0.1y(\infty)$ to $0.9y(\infty)$
- overshoot M_p and peak time t_p
- settling time t_s — first time for transients to decay to within a specified small percentage of $y(\infty)$ and stay in that range (we will usually worry about 5% settling time)

Transient-Response (or Time-Domain) Specs



Do we want these quantities to be large or small?

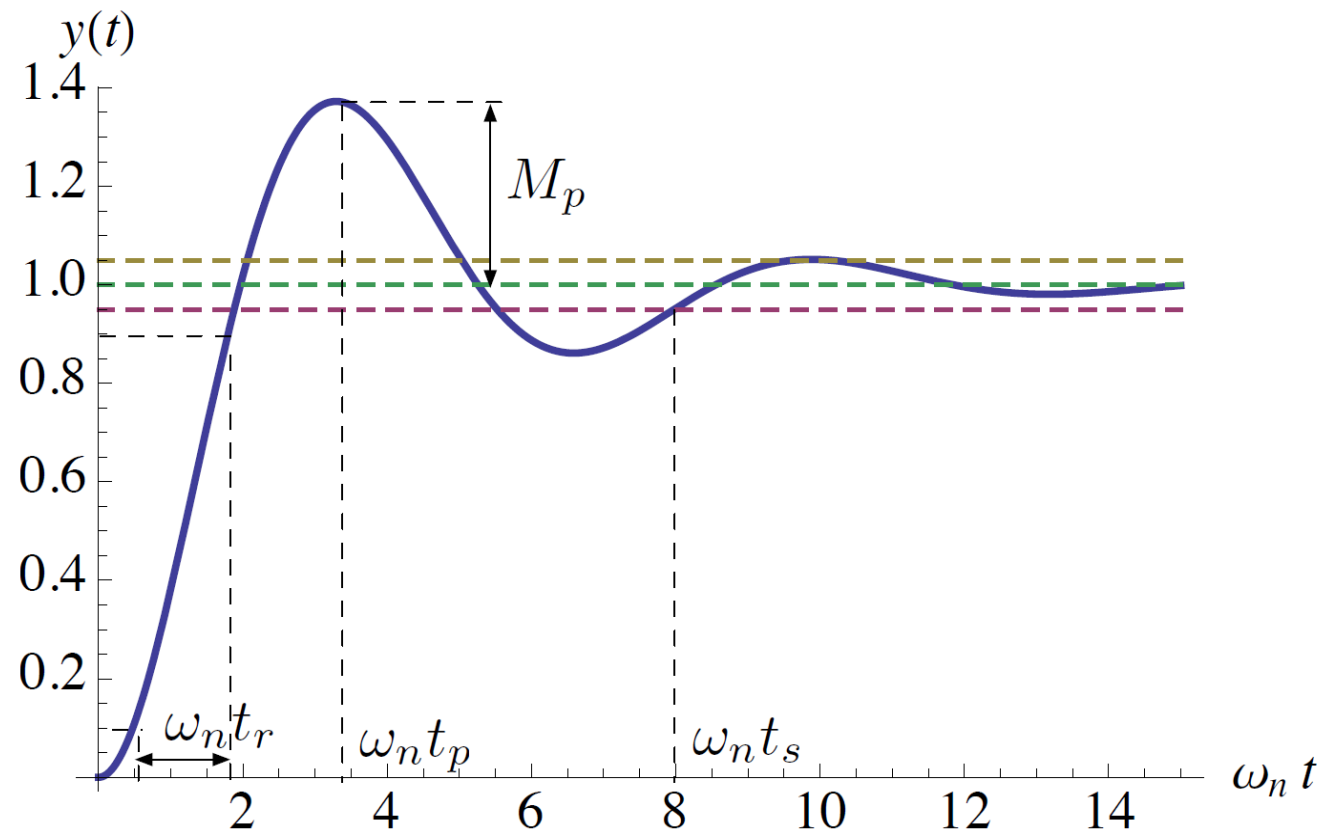
Transient-Response (or Time-Domain) Specs



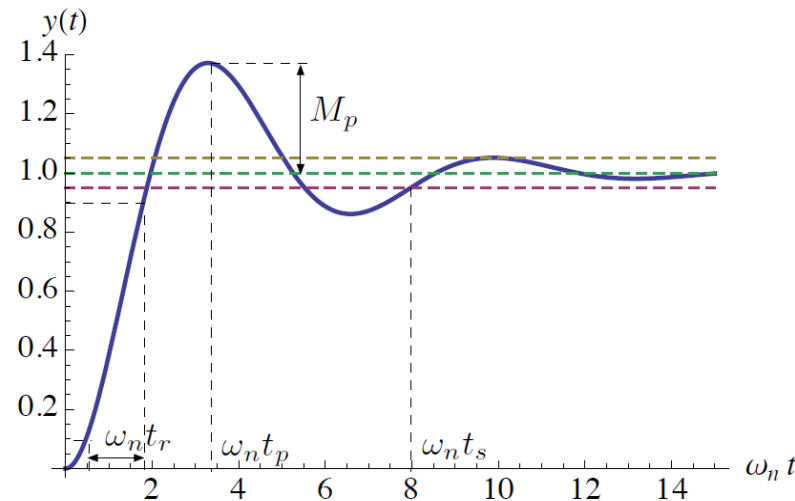
Do we want these quantities to be large or small?

- ▶ t_r small
- ▶ M_p small
- ▶ t_p small
- ▶ t_s small

Trade-offs among specs: decrease $t_r \longrightarrow$ increase M_p , etc.



Formulas for TD Specs: Rise Time



Empirical
result!

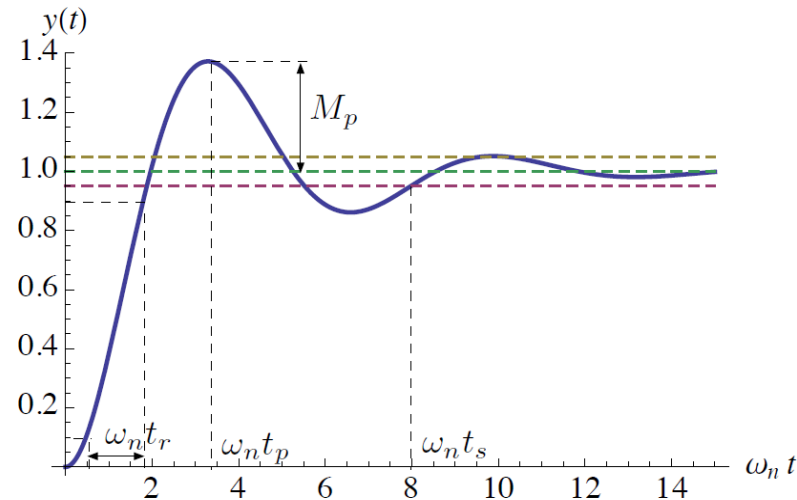
Rise time t_r — hard to calculate analytically.

Empirically, on the normalized time scale ($t \rightarrow \omega_n t$), rise times are *approximately* the same

$$\omega_n t_r \approx 1.8 \quad (\text{exact for } \zeta = 0.5)$$

So, we will work with $t_r \approx \frac{1.8}{\omega_n}$ (good approx. when $\zeta \approx 0.5$)

Formulas for TD Specs: Overshoot & Peak Time



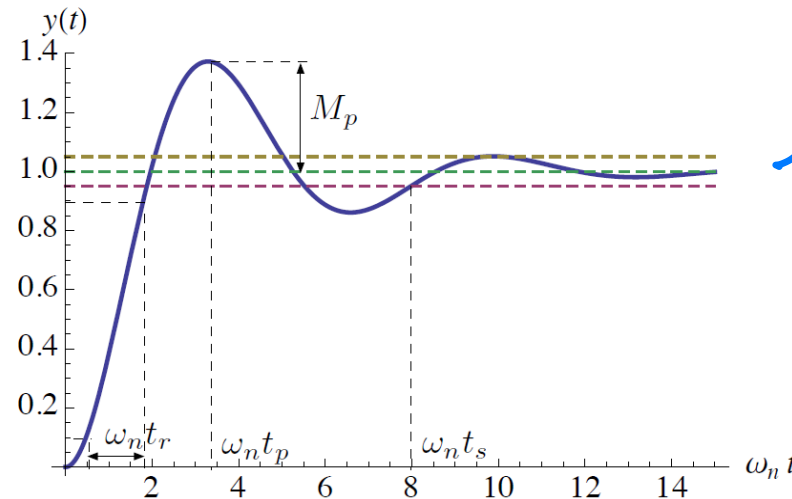
t_p is the *first time* $t > 0$ when $y'(t) = 0$

$$y(t) = 1 - e^{-\sigma t} \left(\cos(\omega_d t) + \frac{\sigma}{\omega_d} \sin(\omega_d t) \right)$$

$$y'(t) = \left(\frac{\sigma^2}{\omega_d} + \omega_d \right) e^{-\sigma t} \sin(\omega_d t) = 0 \text{ when } \omega_d t = 0, \pi, 2\pi, \dots$$

so $t_p = \frac{\pi}{\omega_d}$

Formulas for TD Specs: Overshoot & Peak Time



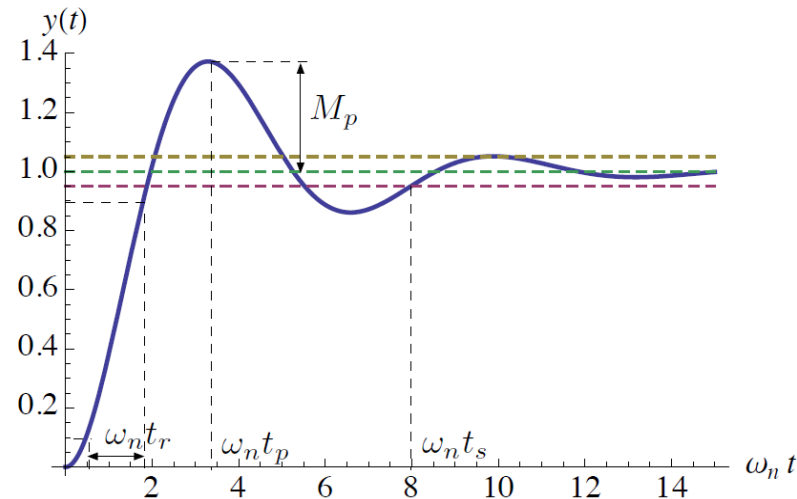
first $y'(t) = 0$
and $t > 0$

We have just computed $t_p = \frac{\pi}{\omega_d}$

To find M_p , plug this value into $y(t)$:

$$\begin{aligned} M_p &= y(t_p) - 1 = -e^{-\frac{\sigma\pi}{\omega_d}} \left(\cos \left(\omega_d \frac{\pi}{\omega_d} \right) + \frac{\sigma}{\omega_d} \sin \left(\omega_d \frac{\pi}{\omega_d} \right) \right) \\ &= \exp \left(-\frac{\sigma\pi}{\omega_d} \right) = \exp \left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}} \right) \quad \text{--- exact formula} \end{aligned}$$

Formulas for TD Specs: Settling Time



$$t_s = \min \left\{ t > 0 : \frac{|y(t') - y(\infty)|}{y(\infty)} \leq 0.05 \text{ for all } t' \geq t \right\} \text{ (here, } y(\infty) = 1)$$

$$|y(t) - 1| = e^{-\sigma t} \left| \cos(\omega_d t) + \frac{\sigma}{\omega_d} \sin(\omega_d t) \right|$$

here, $e^{-\sigma t}$ is what matters (sin and cos are bounded between ± 1), so $e^{-\sigma t_s} \leq 0.05$ this gives $t_s = -\frac{\ln 0.05}{\sigma} \approx \frac{3}{\sigma}$

Formulas for TD Specs

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\sigma^2 + \omega_d^2}{(s + \sigma)^2 + \omega_d^2}$$

$s^2 + a_1 s + a_0, \quad \omega_n = \sqrt{a_0}$

$2\zeta\omega_n = a_1$

$\zeta = \frac{a_1}{2\omega_n} = \frac{a_1}{2\sqrt{a_0}}$

$$t_r \approx \frac{1.8}{\omega_n}$$

$$t_p = \frac{\pi}{\omega_d}$$

$$M_p = \exp\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

$$t_s \approx \frac{3}{\sigma}$$

TD Specs in Frequency Domain

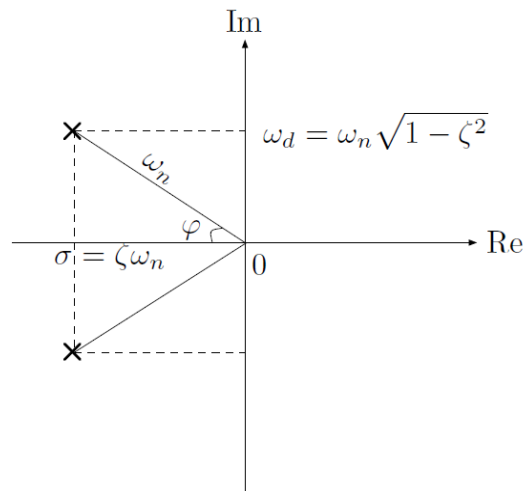
We want to *visualize* time-domain specs in terms of *admissible pole locations* for the 2nd-order system

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\sigma^2 + \omega_d^2}{(s + \sigma)^2 + \omega_d^2}$$

$$\text{where } \sigma = \zeta\omega_n$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\text{Step response: } y(t) = 1 - e^{-\sigma t} \left(\cos(\omega_d t) + \frac{\sigma}{\omega_d} \sin(\omega_d t) \right)$$



$$\omega_n^2 = \sigma^2 + \omega_d^2$$

$$\zeta = \cos \varphi$$

Rise Time in Frequency Domain

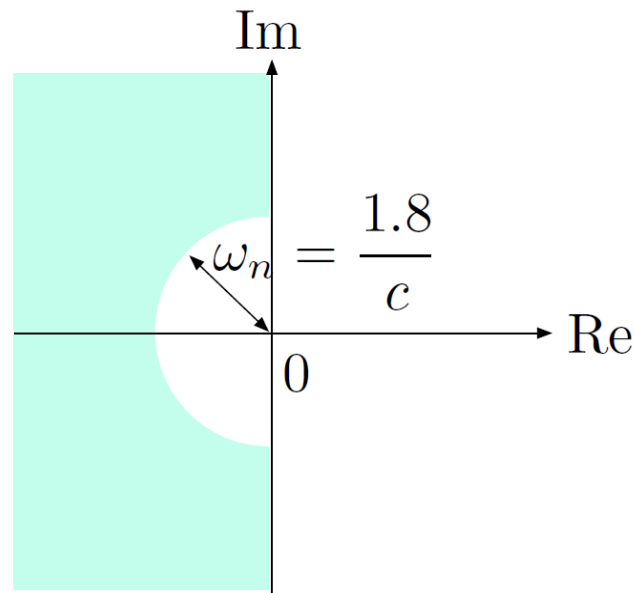
Suppose we want $t_r \leq c$ (c is some desired given value)

Rise Time in Frequency Domain

Suppose we want $t_r \leq c$ (c is some desired given value)

$$t_r \approx \frac{1.8}{\omega_n} \leq c \quad \implies \quad \omega_n \geq \frac{1.8}{c}$$

Geometrically, we want poles to lie in the shaded region:



(recall that ω_n is the *magnitude of the poles*)

Overshoot in Frequency Domain

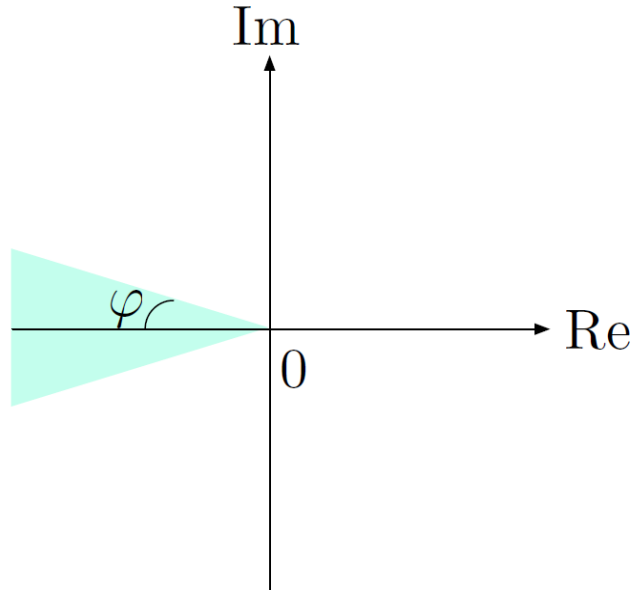
Suppose we want $M_p \leq c$

Overshoot in Frequency Domain

Suppose we want $M_p \leq c$

$$\underbrace{M_p = \exp\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)}_{\text{decreasing function}} \leq c \quad \text{--- need large damping ratio}$$

Geometrically, we want poles to lie in the shaded region:



$$\begin{aligned} \frac{\zeta}{\sqrt{1-\zeta^2}} &= \frac{\omega_n \zeta}{\omega_n \sqrt{1-\zeta^2}} \\ &= \frac{\sigma}{\omega_d} = \cot \varphi \end{aligned}$$

--- need φ to be small

Intuition: good damping $\rightarrow$
good decay in $1/2$ period

Settling Time in Frequency Domain

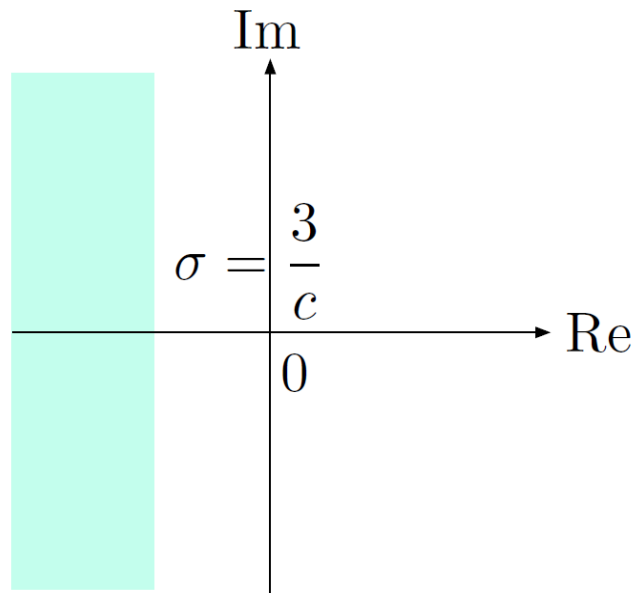
Suppose we want $t_s \leq c$

Settling Time in Frequency Domain

Suppose we want $t_s \leq c$

$$t_s \approx \frac{3}{\sigma} \leq c \quad \implies \quad \sigma \geq \frac{3}{c}$$

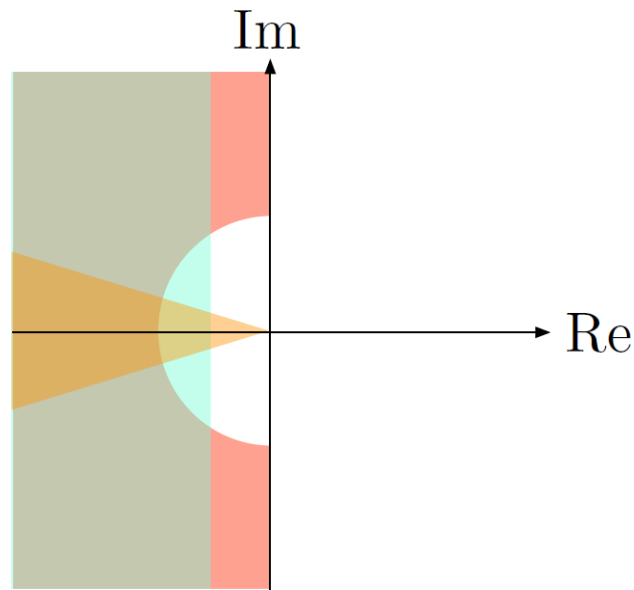
Want poles to be sufficiently fast (large enough magnitude of real part):



Intuition: poles far to the left $\rightarrow$ transients decay faster $\rightarrow$ smaller t_s

Combination of Specs

If we have specs for any combination of t_r , M_p , t_s , we can easily relate them to allowed pole locations:



The shape and size of the region for admissible pole locations will change depending on which specs are more severely constrained.

This is very appealing to engineers: easy to visualize things, no such crisp visualization in time domain.

But: not very rigorous, and also only valid for our prototype 2nd-order system, which has only 2 poles and no zeros ...

Review

- The poles of the 2nd-order system?
- What's damping ratio? How it affects the system?
- What's the frequency? How it affects the system?
- What are the meanings and how to calculate the following response specifications:
 - Rise time
 - Overshoot and peak time
 - Settling time
- How to represent it in the frequency domain?

Appendix

$$\mathcal{L}^{-1} \left\{ \frac{H(s)}{s} \right\} = \mathcal{L}^{-1} \left\{ \frac{\sigma^2 + \omega_d^2}{s[(s + \sigma)^2 + \omega_d^2]} \right\} = 1 - e^{-\sigma t} \left(\cos(\omega_d t) + \frac{\sigma}{\omega_d} \sin(\omega_d t) \right)$$

Proof:

$$\frac{\sigma^2 + \omega_d^2}{s[(s + \sigma)^2 + \omega_d^2]} = \frac{A}{s} + \frac{Bs + C}{(s + \sigma)^2 + \omega_d^2} = \frac{1}{s} + \frac{-s - 2\sigma}{(s + \sigma)^2 + \omega_d^2} \quad (A = 1, B = -1, C = -2\sigma)$$

$$\mathcal{L}^{-1} \left(\frac{\sigma^2 + \omega_d^2}{s[(s + \sigma)^2 + \omega_d^2]} \right) = \mathcal{L}^{-1} \left(\frac{1}{s} - \frac{s + \sigma}{(s + \sigma)^2 + \omega_d^2} - \frac{\sigma}{(s + \sigma)^2 + \omega_d^2} \right)$$

$$= \sigma(t) - e^{-\sigma t} \cos \omega_d t - e^{-\sigma t} \frac{\sigma}{\omega_d} \sin \omega_d t$$

$$= \sigma(t) - e^{-\sigma t} \left(\cos \omega_d t + \frac{\sigma}{\omega_d} \sin \omega_d t \right)$$

$$= 1 - e^{-\sigma t} \left(\cos \omega_d t + \frac{\sigma}{\omega_d} \sin \omega_d t \right)$$